HISTOLOGY OF TONGUE

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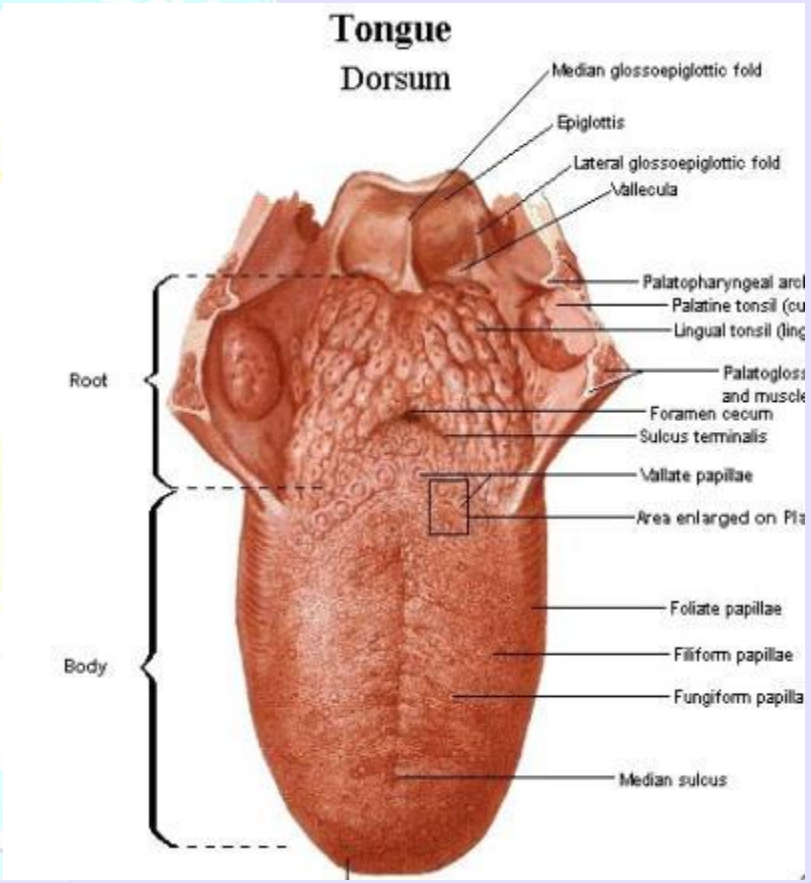
“LEARNING OBJECTIVE”

At the end of lecture student must be able :

- To identify the epithelium lining its mucosa.
- To know the structure of Taste buds
- To define structure of papillae
- To know different types of papillae.
- To explain glands present in the tongue and their microscopic structure

TONGUE

- It is a mass of skeletal muscle covered on both side with mucosa.
- Muscles fibers in the tongue run in three directions, longitudinal, transverse and vertical.
- The mucosa in the anterior part is rough due to the presence of projections called papilla.
- In the posterior part the mucosa contains lymphoid follicle.

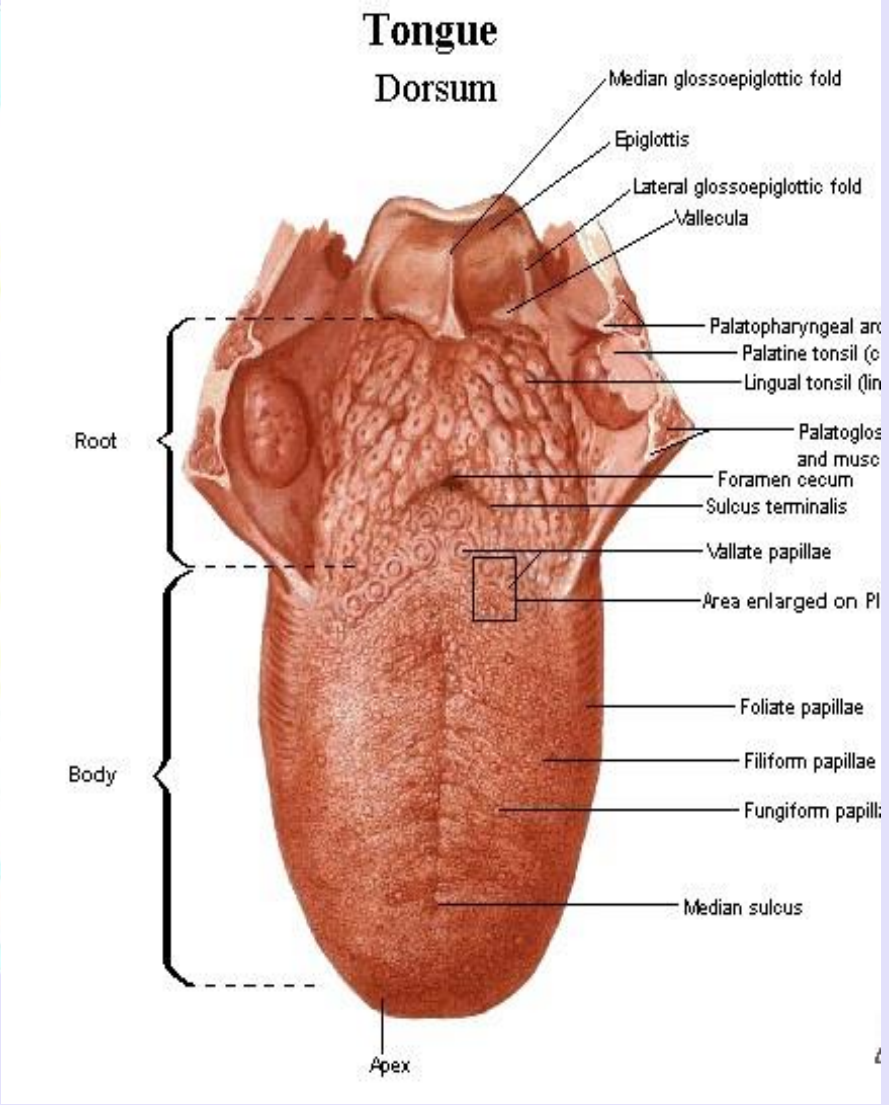


TONGUE

- The dorsal surface of the tongue is divided by a V-shaped sulcus terminalis into:
- 1- An oral part, the anterior two-thirds.
- 2- A pharyngeal part, the posterior one-third.

- The dorsal surface of the oral part has the lingual papillae.

- The lingual papillae consist of a connective tissue core covered with stratified squamous epithelium.

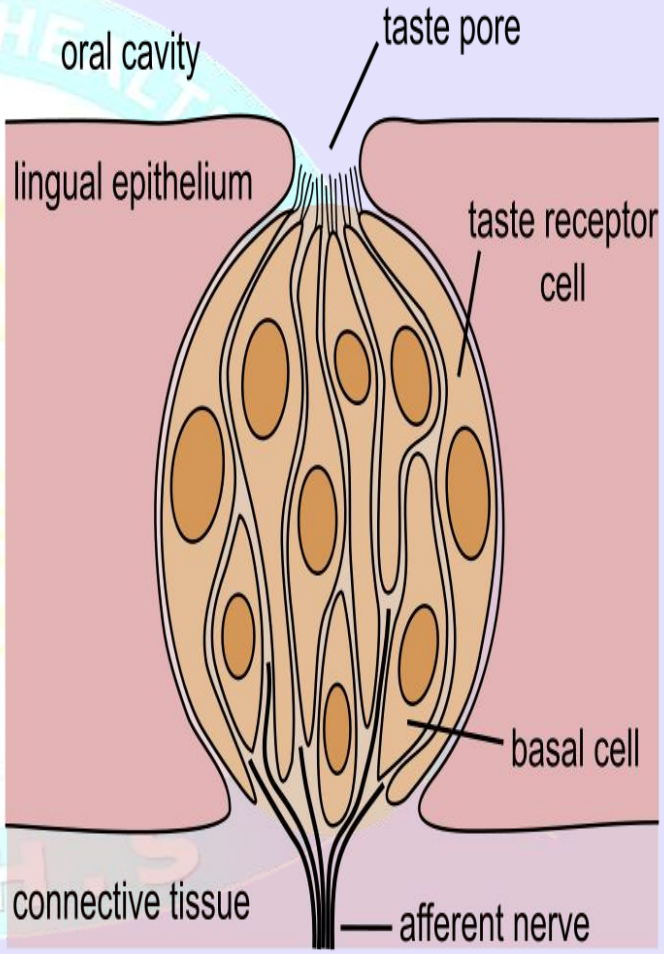


“TASTE BUDS”

- Taste buds contain the receptors for taste.

- They are located around the small structures on the upper surface of the tongue, soft palate, upper esophagus and epiglottis, which are called papillae.

- These structures are involved in detecting the five (known) elements of taste perception: salty, sour, bitter, sweet, and savory (or umami).



TASTE BUDS

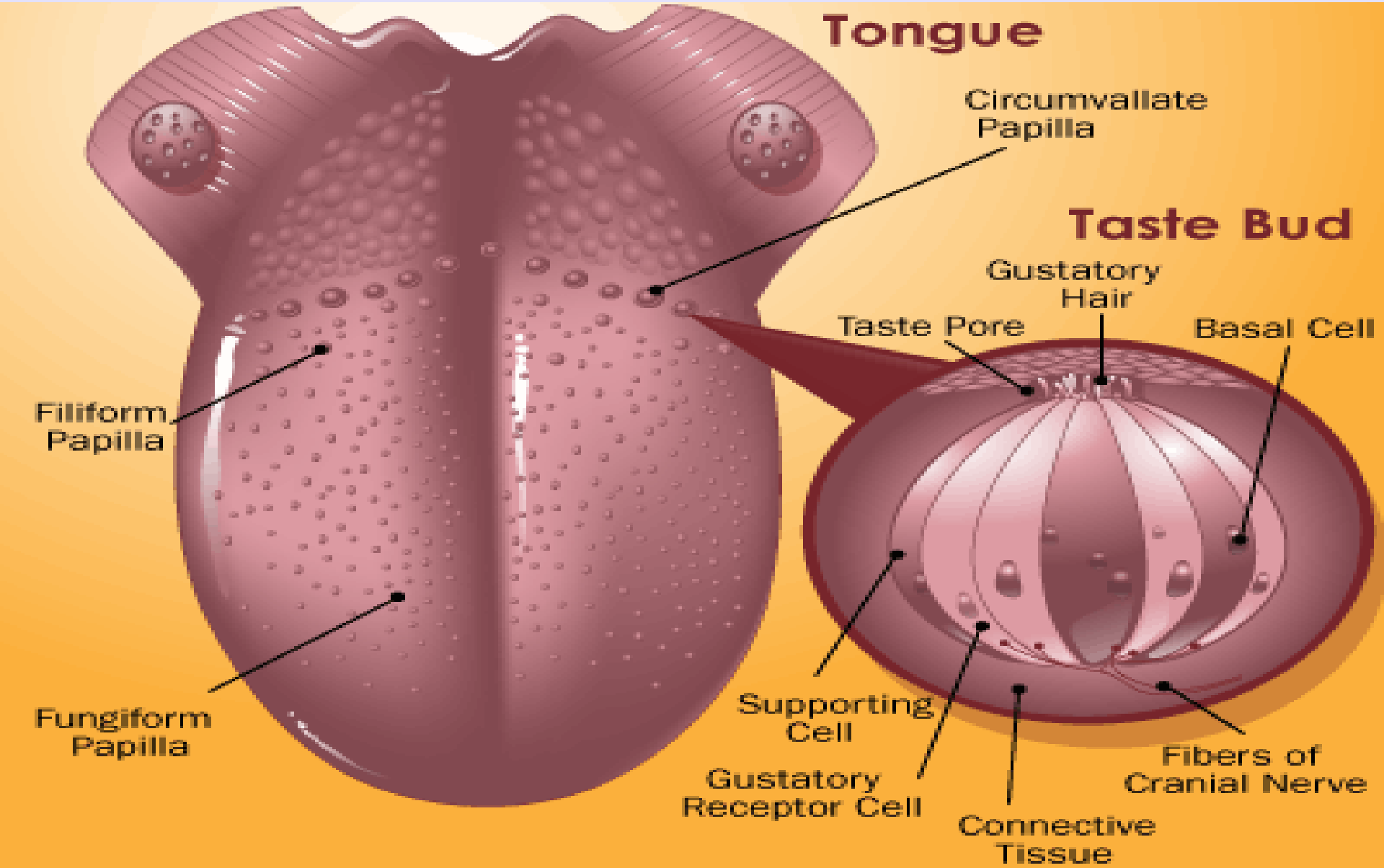
- Via small openings in the tongue epithelium, called taste pores, parts of the food dissolved in saliva come into contact with taste receptors.
- These are located on top of the taste receptor cells that constitute the taste buds.
- The taste receptor cells send information detected by clusters of various receptors and ion channels to the gustatory areas of the brain via the seventh, ninth and tenth cranial nerves.
- On average, the human tongue has 2,000–8,000 taste buds.

- The elongated cells that form the taste bud can functionally be divided into three groups:

1. Sensory cells (gustatory cells)
2. Supporting (or sustentacular) cells,
3. Basal cells.

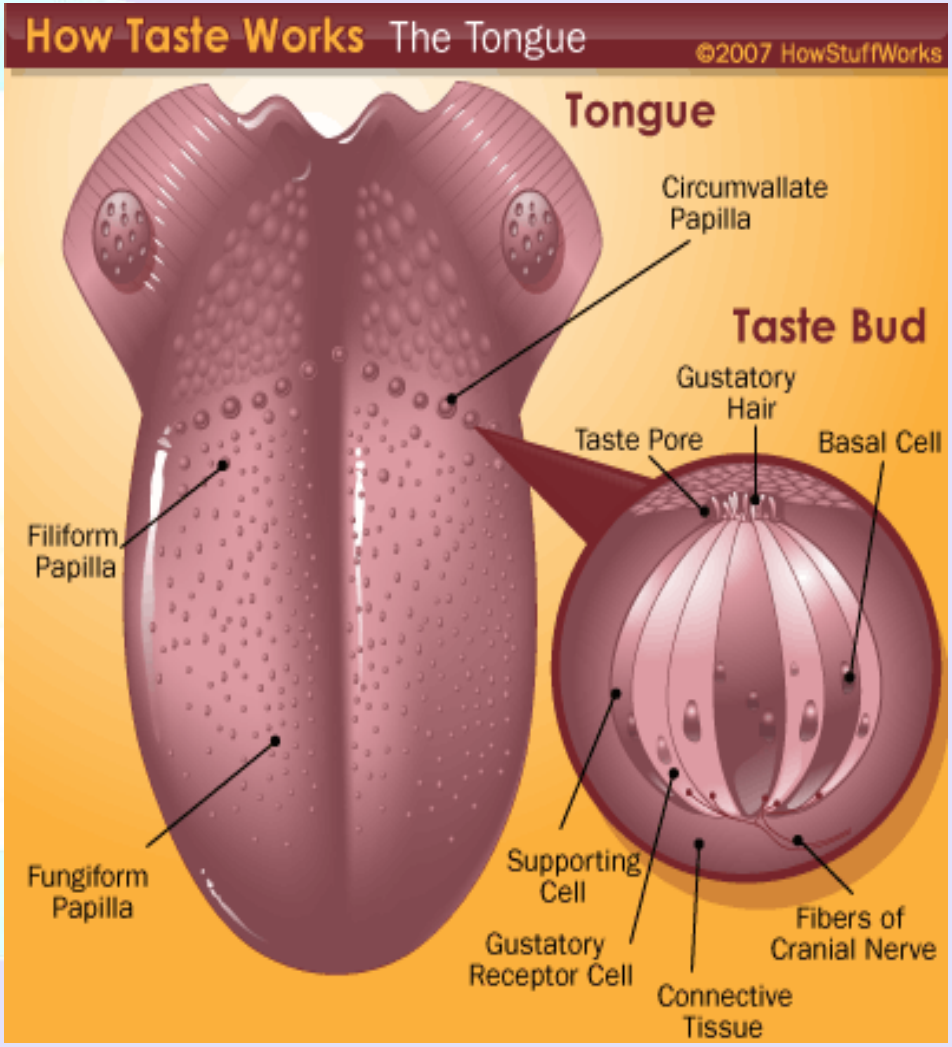
- Sensory cells extend microvilli into the taste pore.
- The microvilli contain the receptors for the different basic taste modalities (sweet, salty, bitter and acid).
- Basal cells regenerate the two other cell types

“TASTE BUDS”


Tongue
Circumvallate Papilla
Taste Bud
Gustatory Hair
Taste Pore
Basal Cell
Filiform Papilla
Fungiform Papilla
Supporting Cell
Gustatory Receptor Cell
Connective Tissue
Fibers of Cranial Nerve

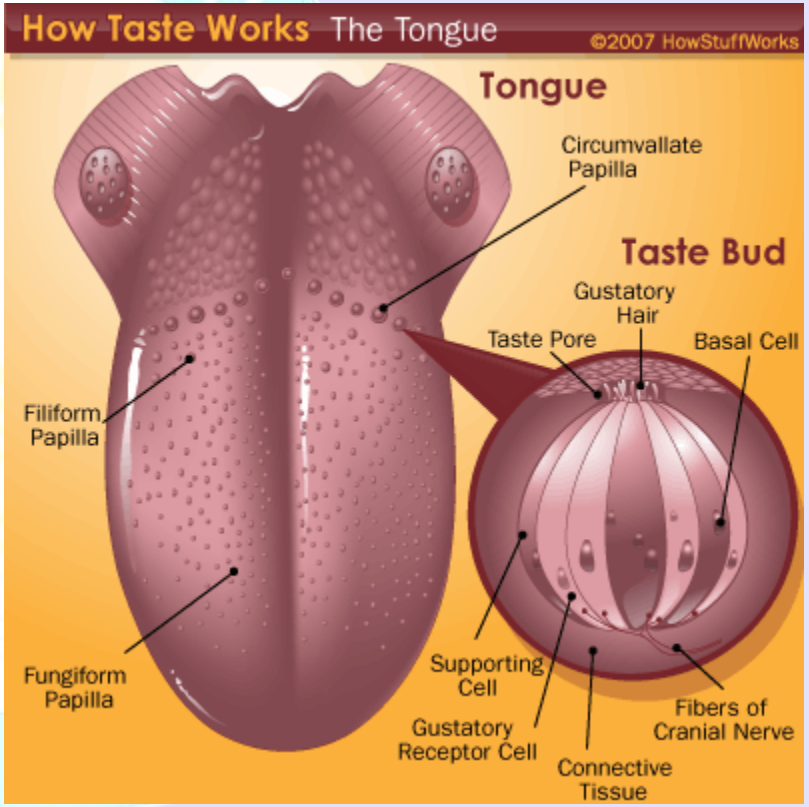
STRUCTURE OF TASTE BUDS

- Each taste bud is flask-like in shape, its broad base resting on the corium, and its neck opening, the gustatory pore, between the cells of the epithelium.
- The bud is formed by two kinds of cells:
- #supporting cells.
- #gustatory cells.



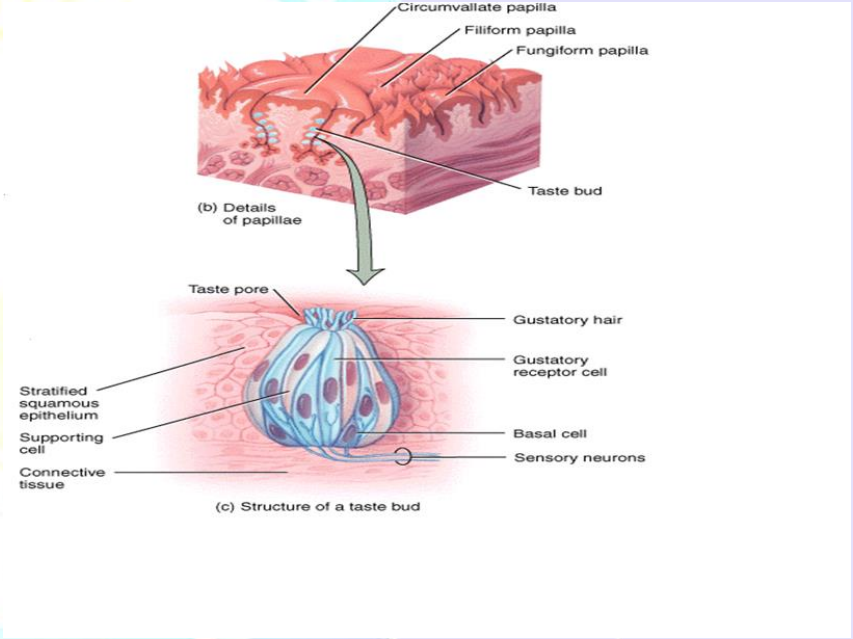
The supporting (sustentacular) cells

- Are mostly arranged like the staves of a cask.
- Form an outer envelope for the bud.
- Some, however, are found in the interior of the bud between the gustatory cells.



The gustatory (taste) cells

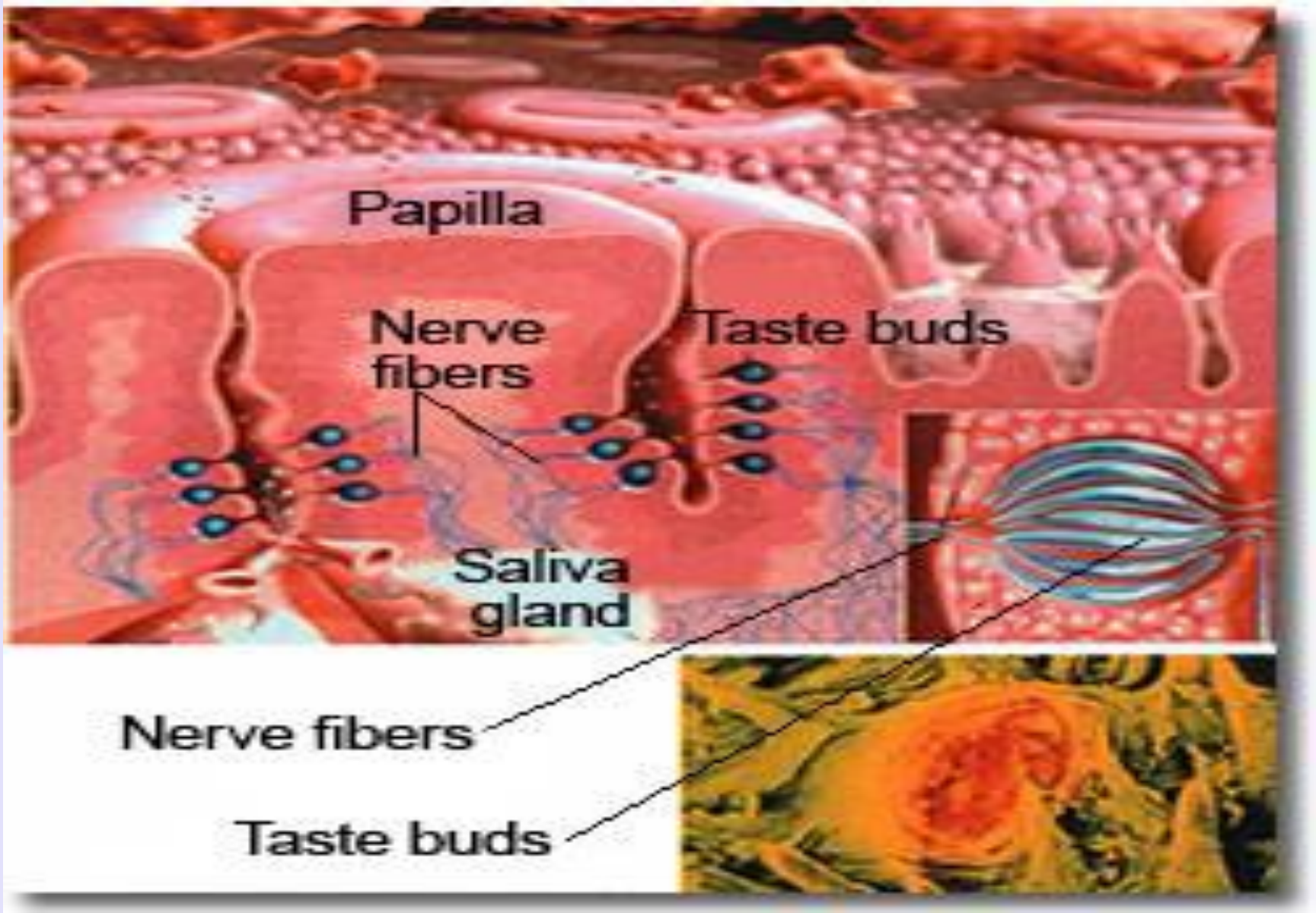
- The gustatory (taste) cells, a chemoreceptor.
- Occupy the central portion of the bud.
- They are spindle-shaped.
- Each possesses a large spherical nucleus near the middle of the cell.



STRUCTURE OF TASTE BUDS

- The central process passes toward the deep extremity of the bud, and there ends in single or bifurcated varicosities.
- The nerve fibrils after losing their medullary sheaths enter the taste bud, and end in fine extremities between the gustatory cells.
- 0ther nerve fibrils ramify between the supporting cells and terminate in fine extremities.
- these, however, are believed to be nerves of ordinary sensation and not gustatory.
- The peripheral end of the cell terminates at the gustatory pore in a fine hair filament, the gustatory hair.

STRUCTURE OF TASTE BUDS

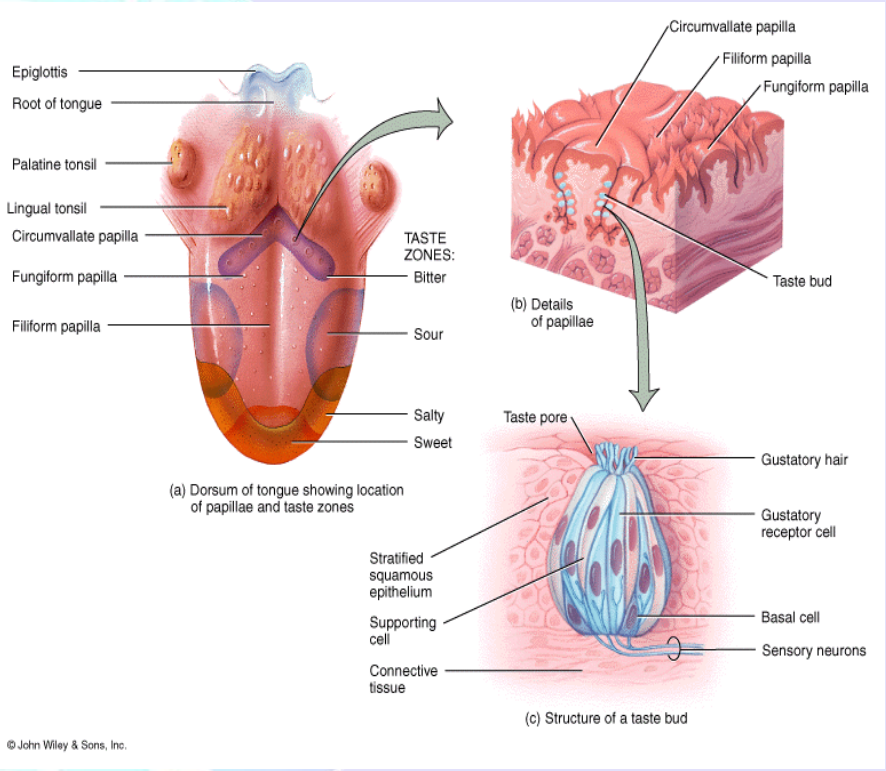


TYPES OF PAPILLAE

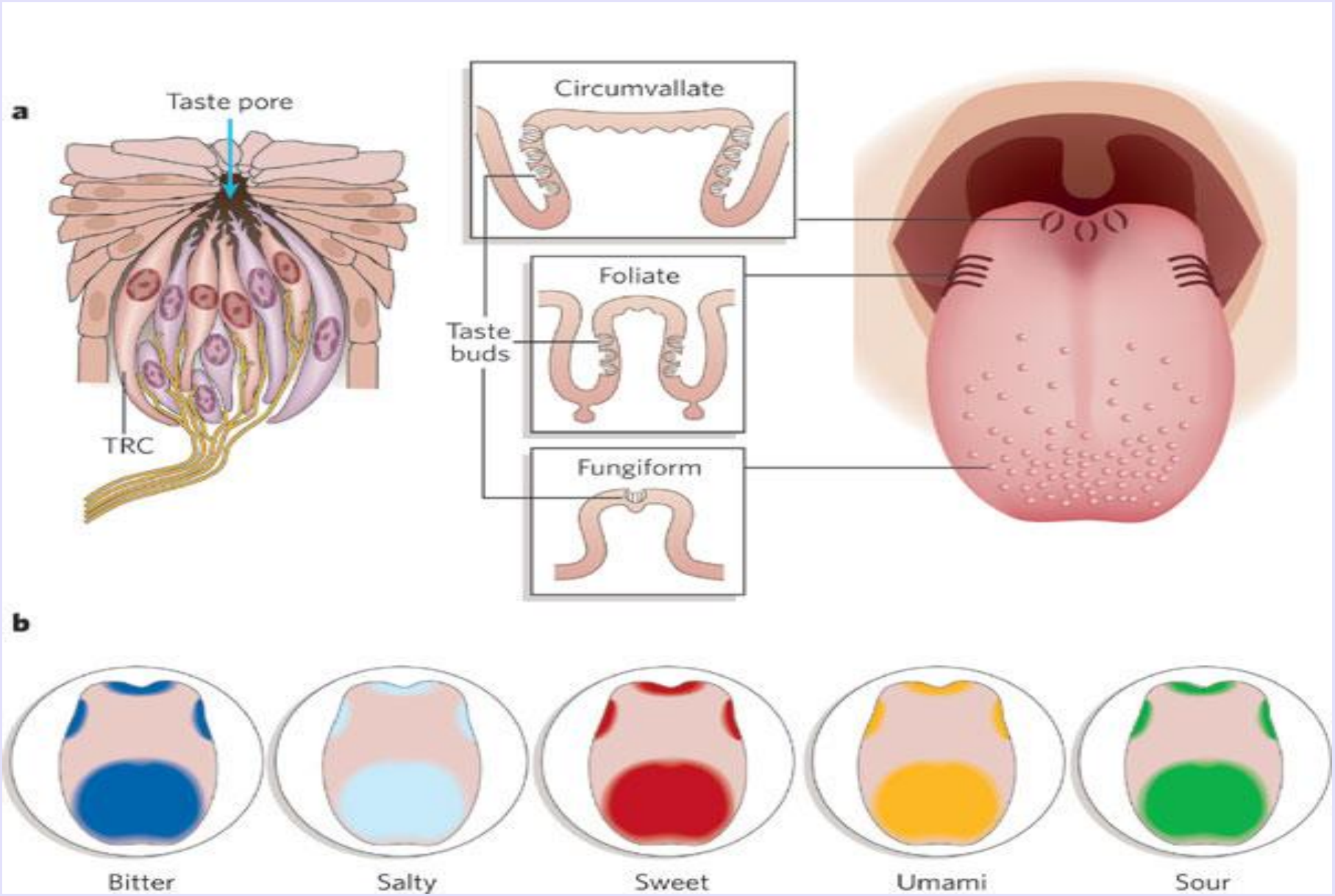
- The majority of taste buds on the tongue sit on raised protrusions of the tongue surface called papillae.

- There are four types of papillae present in the human tongue.

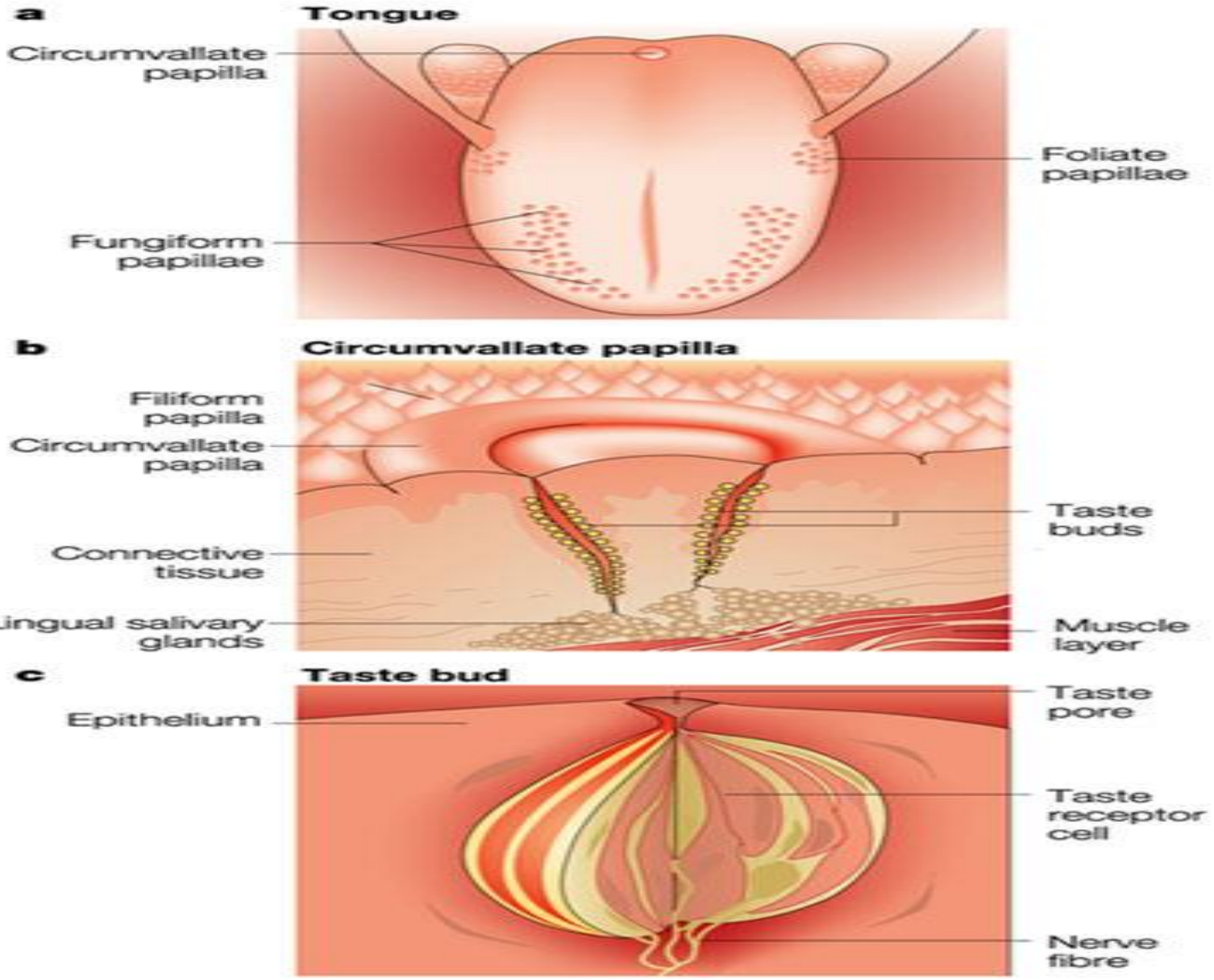
- Fungiform papillae,
- Filiform papillae,
- Foliate papillae,
- Circumvallate papillae.



TYPES OF PAPILLAE



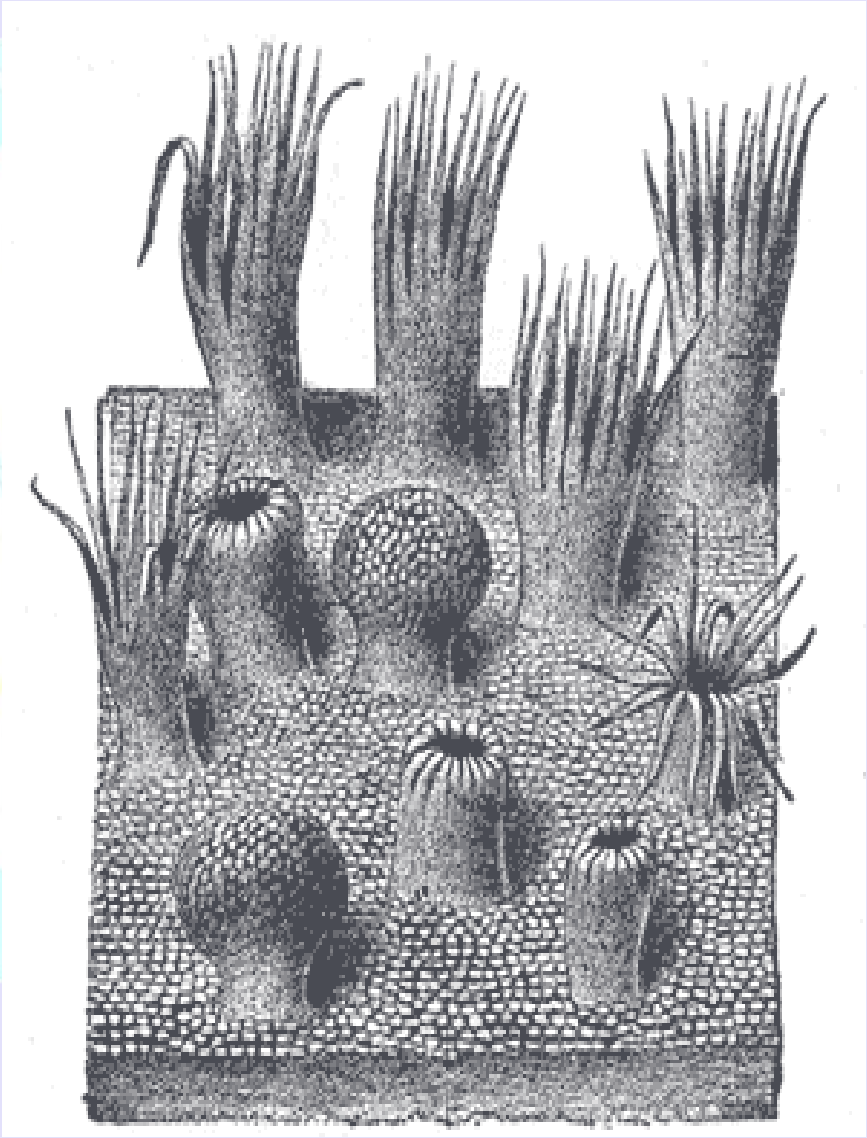
TYPES OF PAPILLAE



TYPES OF PAPILLAE

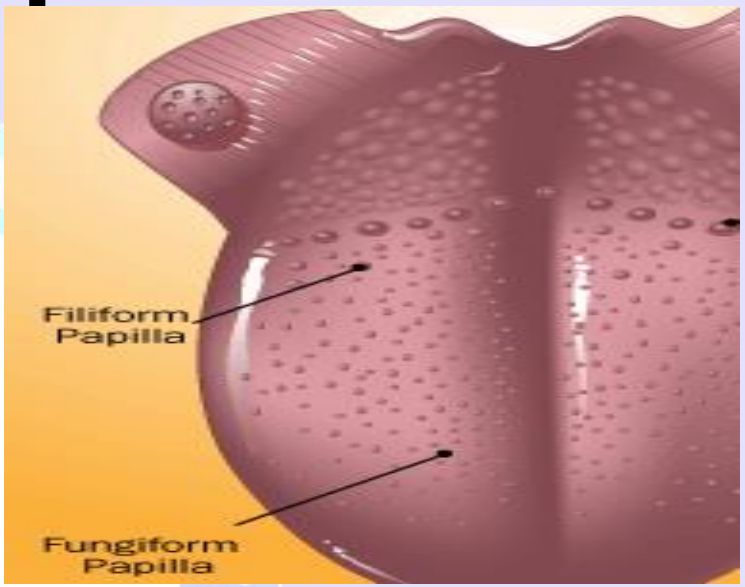
View of a portion of the mucous membrane of the tongue.

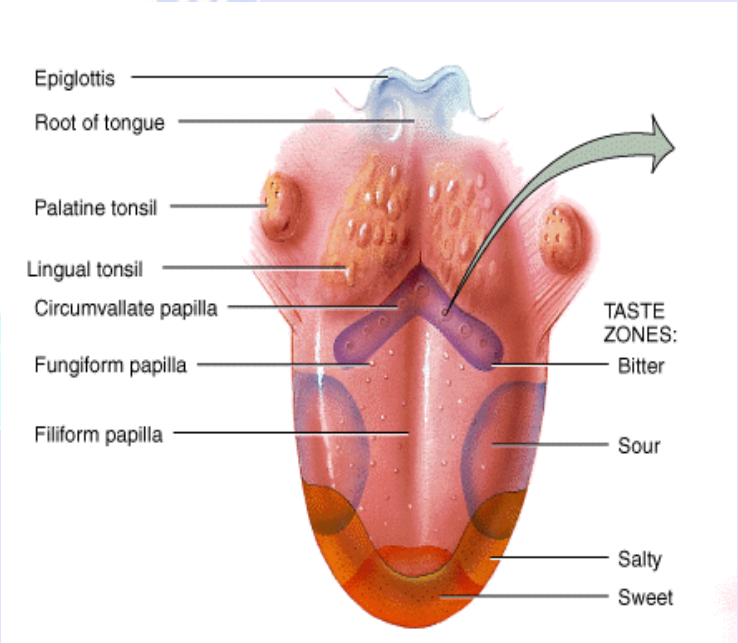
- Two fungiform papillæ are shown.
- On some of the filiform papillae the epithelial prolongations stand erect, in one they are spread out, and in three they are folded in.



Fungiform papillae

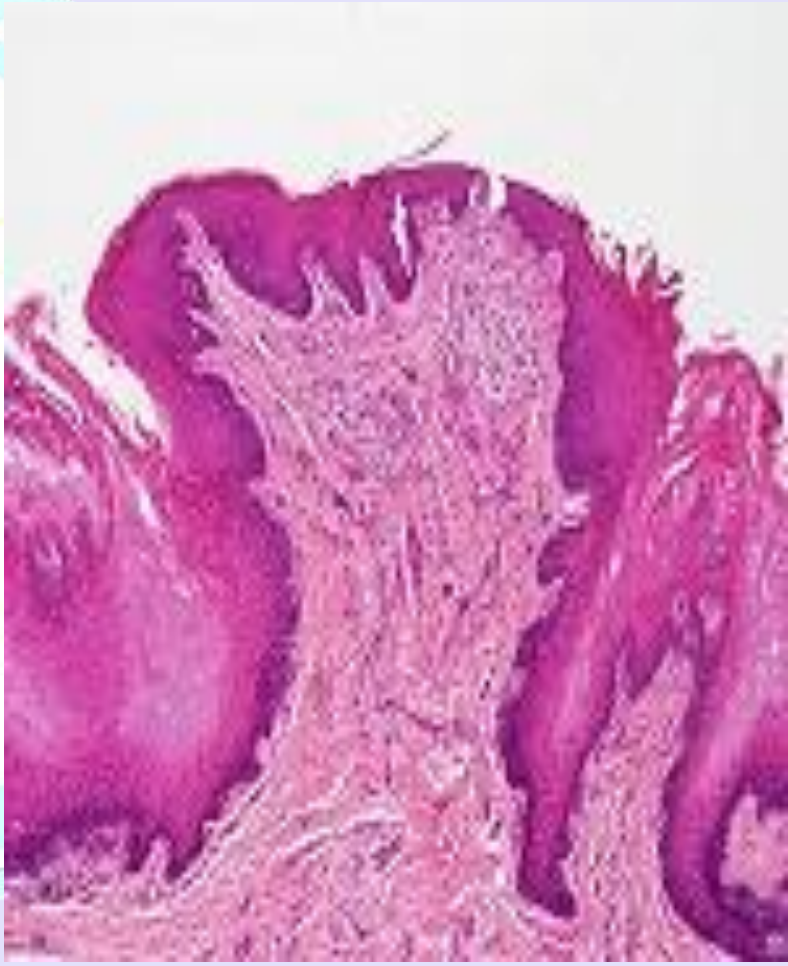
- These are slightly mushroom-shaped if looked at in longitudinal section.
- These are present mostly at the apex (tip) of the tongue, as well as at the sides.
- Innervated by facial nerve.





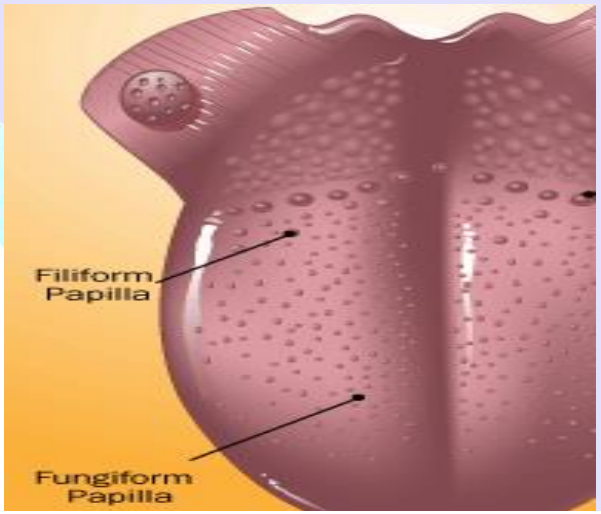
Fungiform Papillae

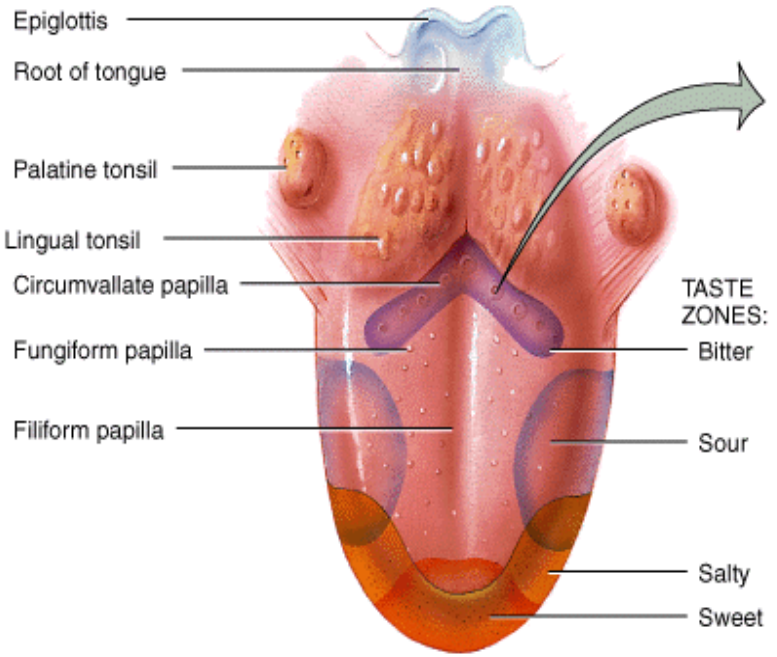
- Fungiform papillae occur singly.
- They are most numerous on the tip of the tongue.
- Central core is made up of vascular connective tissue.
- They are covered by stratified squamous non-cornified epithelium.



Filiform papillae

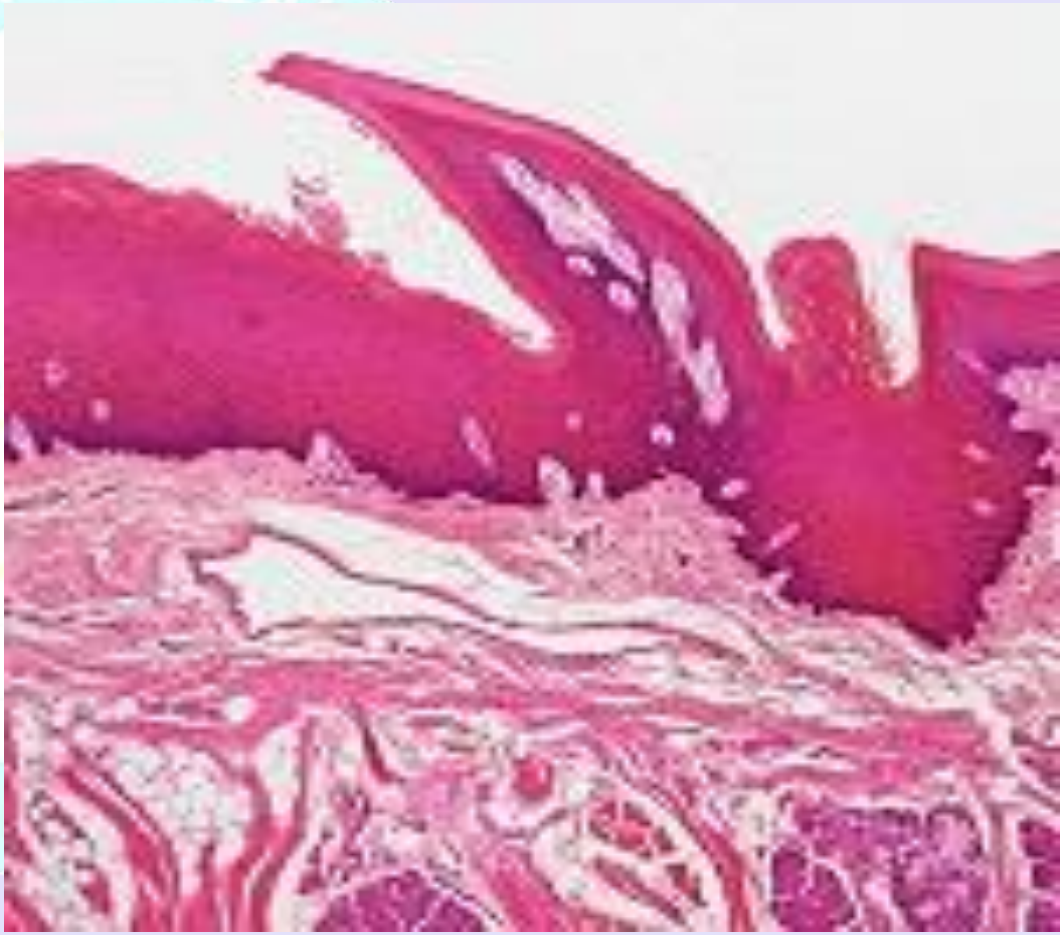
- These are thin.
- Long papillae .
- "V"-shaped cones
- That don't contain taste buds but are the most numerous.
- These papillae are mechanical and not involved in gustation.
- They are characterized by increased keratinization.



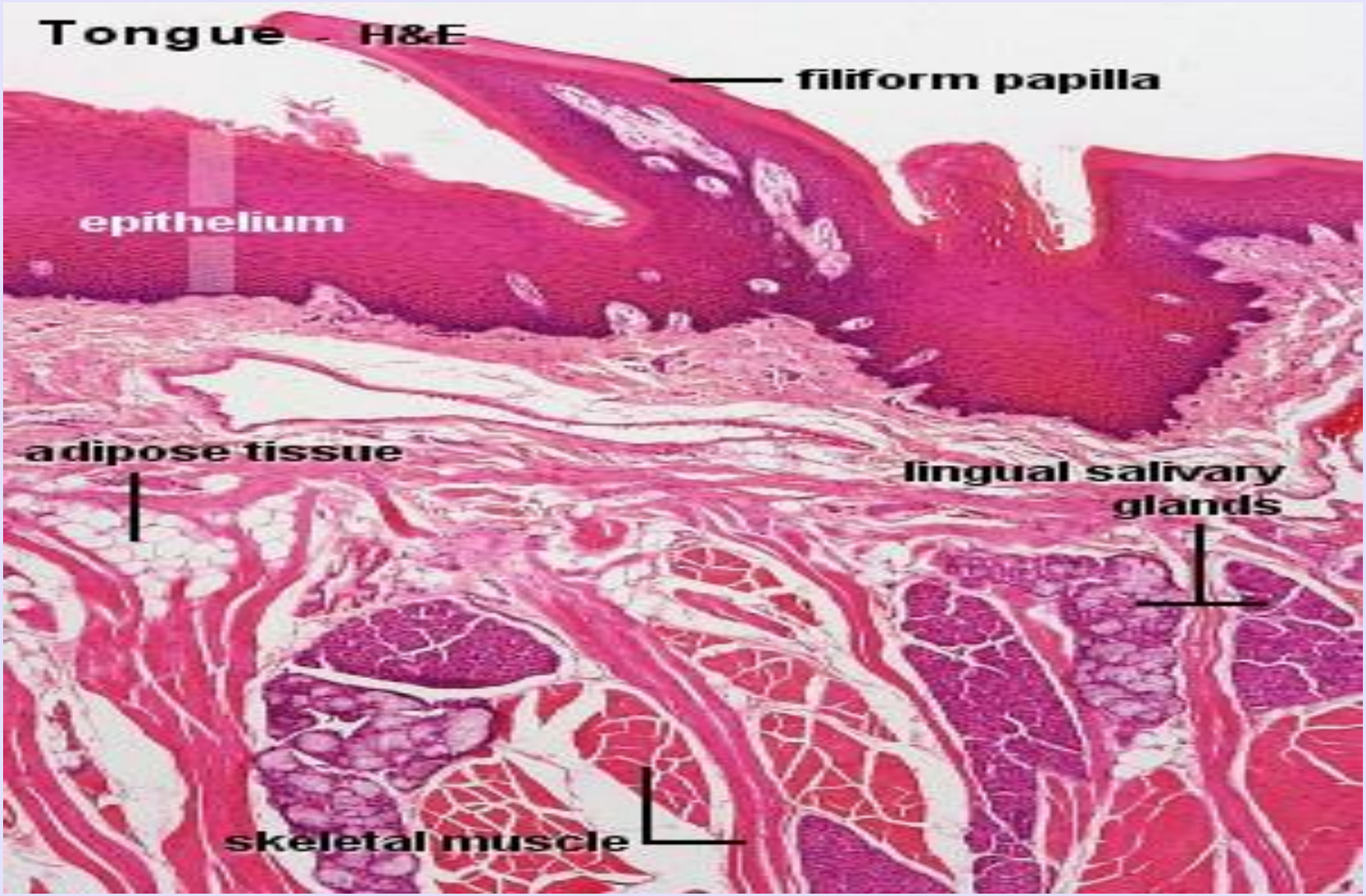


Filiform Papillae

- Filiform papillae are the smallest and most numerous papillae.
- They are thread like, covered by stratified squamous cornified epithelium.
- They aid in the manipulation and processing of foods



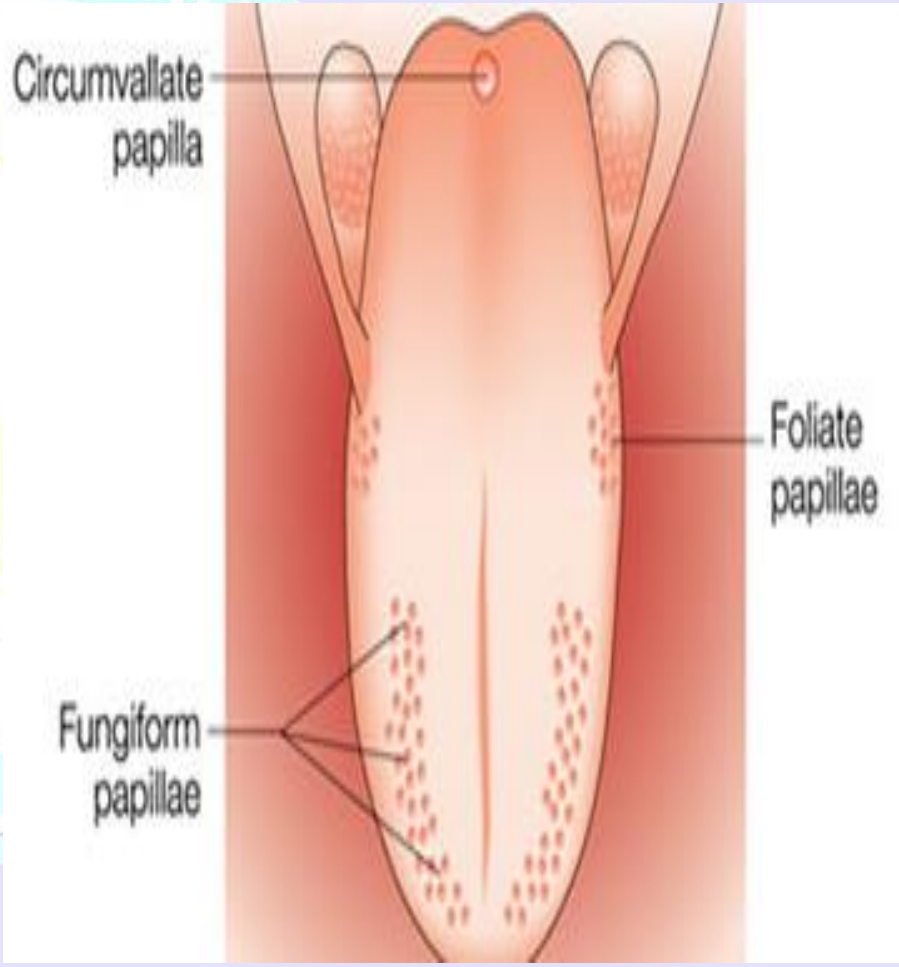
Filiform papillae



Foliate papillae

- These are ridges and grooves towards the posterior part of the tongue found on lateral margins.
- Innervated by,

  facial nerve (anterior papillae)
- Glossopharyngeal nerve (posterior papillae).



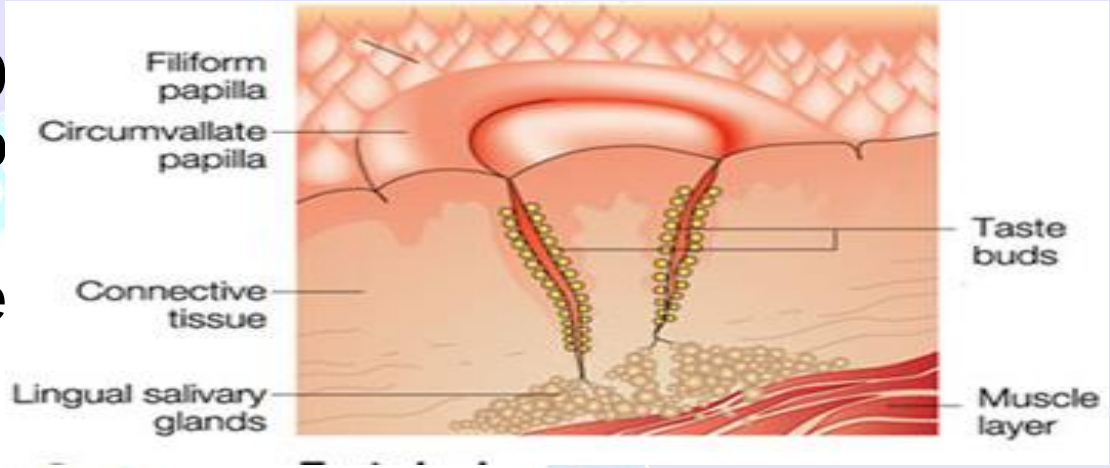
Foliate Papilla

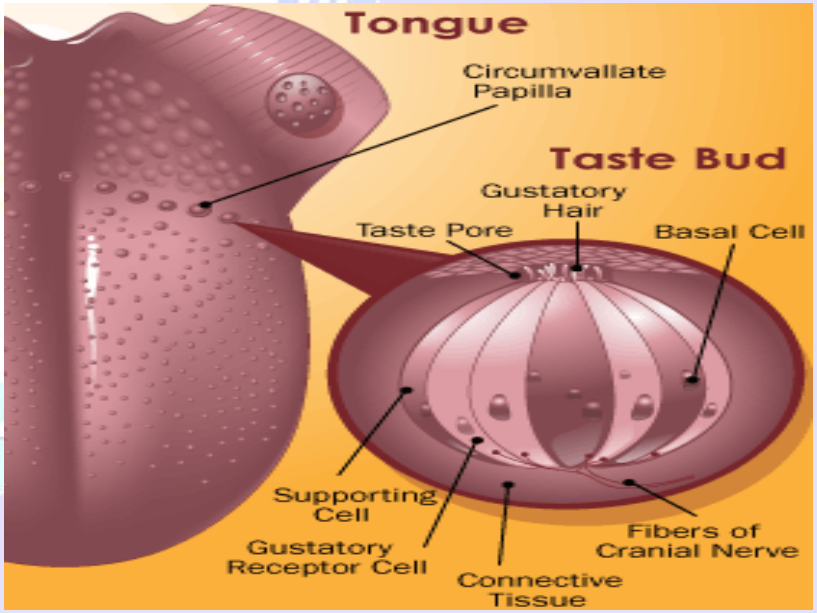
- Foliate papilla are not well developed in humans and may be absent in aged individuals.

- If present, they form lamellae along the posterior and lateral borders of the tongue

Circumvallate papillae

- There are only about 10 to 14 of these papillae o most people.
- They are present at the back of the oral part of the tongue.
- They are arranged in a circular-shaped row just in front of the sulcus terminalis of the tongue.
- They are associated with ducts of **Von Ebner's** glands.
- Are innervated by the glossopharyngeal nerve.





Circumvallate Papillae

- Circumvallate papillae are the largest and least numerous papillae.
- In humans there are between 8 and 12 of them.
- They occur in depressions of the surface of the tongue and are surrounded with a trench formed by the infolding of the epithelium.
- Taste buds are particularly numerous on the lateral surfaces of these papillae



GLANDS OF THE TONGUE

- Many serous and mucous salivary glands are scattered throughout the muscle and lamina propria.
- The tongue glands occur in three groups.

1- Anterior lingual glands (mixed type)

2- Glands of von Ebner (serous type)

3- Mucous glands of the root

NERVE SUPPLY

- A number of the cranial nerves supply the tongue.

Anterior two-thirds

- V, trigeminal nerve - sensory
- VII, facial nerve - taste

Posterior one-third

- IX, glossopharyngeal nerve - sensory/taste.

Motor

- XII, hypoglossal nerve.

FUNCTION OF THE TONGUE

- Manipulation of food
- General sensory reception
- Special sense (taste sensation)



Thanks